Round 1 :
1. What is Merge and Append Queries?
2. What is difference between Star Schema and Snowflake Schema?
3. What are the data connections available in Power BI?
4. What are the types of Gateway in Power BI and Explain it in detail?
5. What is Sync slicer? And how will you incorporate it if have multiple pages and how will you filter a data?
6. What is RLS and Explain it in detail that how will you achieve it?
7. What is Drill Through concepts in Power BI?
8. What are the charts used in real time project and explain some of the charts?
9. What are the charts used in real time project and Explain some of the charts?
10. Explain in detail about the Decomposition tree chart with Proper Example?
11. Explain about the types of DAX functions in Power BI and Tell me about the time intelligence Function?
12. If I have a table with date column, I want to calculate month wise sales how will you achieve and what will be your approach?
13. How to handle many to many relationship in data modelling in Power BI?
14. What are the joins available in SQL and Explain in detail about each joins?
15. What are DDL and DML commands in SQL?

Round 2 :
1. How to convert the number into date format. The number which we need to convert is 20240629. How will you convert this?
2. We have a name column in a table which contains first and last name (Benjamin-Smith). How we can get only the Last Name from existing column through DAX function?
3. What are the possible ways of calculating Division in DAX?
4. How can we extract only the numbers from below mentioned column and create a new column?
5. Display the colour only for the highest sales in the visual through DAX function?
6. How to optimize your report if we have huge data?
7. How do you know if your client have viewed your report or not?
8. If we have two fact tables, which don't share any relationship between those tables, but we need to get the DISTINCT Count of Customer ID from those tables, how to achieve it in DAX?
9. How did you validate your report?
10. What is summarize DAX function?
11. If we have 10 developer and 500 users. For 500 users they need to see the report, what we can do?
12. If we have inactive relationship between the table, how we can connect those tables?
13. If we more than 500 millions rows of data how will you refresh this data if we are using import mode to connect the data?
14. Is the below measure is correct? can we write the measure like this?
Calculate([Total Sales],[Total Sales] > 4000)?
15. Can you tell us the limitation of Direct query mode?

ANSWERS:

Round 1 Answers

1. What is Merge and Append Queries in Power BI?

Merge Queries

Merge is used to combine two tables horizontally based on a common column (similar to SQL JOIN).

Example:

- Orders Table
- Customer Table
- Common Column = CustomerID

After merge, customer details become part of the orders table.

Types of Merge:

- Left Outer Join
- Right Outer Join
- Inner Join
- Full Outer Join
- Anti Joins

Use Case:
Adding customer names, regions, categories into transaction data.

Append Queries

Append is used to combine tables vertically (stack rows).

Example:

- Sales_2024
- Sales_2025

Append creates one large table with all rows.

Use Case:
Combining monthly/yearly files.

---

2. Difference Between Star Schema and Snowflake Schema

Feature	Star Schema	Snowflake Schema
Structure	Simple	Complex
Dimension Tables	Denormalized	Normalized
Performance	Faster	Slower
Joins	Fewer joins	More joins
Storage	More storage	Less storage
Maintenance	Easy	Difficult

Star Schema

- One Fact table
- Multiple Dimension tables
- Most preferred in Power BI

Example:
FactSales connected to:

- DimCustomer
- DimProduct
- DimDate

Snowflake Schema

Dimension tables further split into sub-dimensions.

Example:
Product → Category → Department

Used when normalization is required.

---

3. Data Connections Available in Power BI

Main connection modes:

1. Import Mode
- Data stored inside Power BI
- Fast performance
- Scheduled refresh required
2. Direct Query
- Data remains in source
- Real-time querying
- Slower than import
3. Live Connection
- Direct connection to Analysis Services
- No data storage in PBIX
4. Composite Model
- Combination of Import + DirectQuery

---

4. Types of Gateway in Power BI

1. Personal Gateway

- Single-user gateway
- Used for personal refresh
- Cannot be shared

Use Case:
Developer testing.

---

2. Standard/Enterprise Gateway

- Multi-user gateway
- Centralized

- Supports scheduled refresh + live connection

Use Case:
Enterprise production reports.

Gateway Purpose

Power BI Service cannot directly access on-premises databases.

Gateway acts as a bridge between:

- Power BI Cloud
- On-premise Data Source

---

5. What is Sync Slicer?

Sync slicer synchronizes filters across multiple report pages.

Example:
If Region = Karachi selected on Page 1,
same filter automatically applies on Page 2.

Steps:

1. Create slicer
2. Open View → Sync Slicers
3. Select pages
4. Enable:

- Sync
- Visible

Use Case

- Multi-page dashboards
- Consistent filtering

6. What is RLS (Row Level Security)?

RLS restricts data access for users.

Example:

- Karachi manager sees only Karachi sales
- Lahore manager sees only Lahore sales

Types of RLS

Static RLS

Manual role creation.

Example:
Region = "Karachi"

Dynamic RLS

Uses USERPRINCIPALNAME()

Example:
User email mapped to region dynamically.

Steps to Implement

1. Create Roles
2. Add DAX filter
3. Publish report
4. Assign users in Power BI Service

Example:

[Region] = "Karachi"

Dynamic:

UserTable[Email] = USERPRINCIPALNAME()

---

7. What is Drill Through in Power BI?

Drill Through allows navigation from summary report to detailed report.

Example:
Click on:

- Product Category

Navigate to:

- Detailed product sales page

Steps

1. Create target page
2. Add drill-through field
3. Right click visual
4. Drill Through

---

8 & 9. Charts Used in Real-Time Projects

Common charts:

1. Bar/Column Chart

Used for comparison.

Example:
Sales by Region.

---

2. Line Chart

Used for trends over time.

Example:
Monthly sales growth.

---

3. Pie/Donut Chart

Used for percentage contribution.

Example:
Market share.

---

4. Matrix Table

Detailed tabular analysis.

---

5. KPI Cards

Shows:

- Revenue
- Profit
- Growth %

---

6. Map Visual

Geographical analysis.

---

7. Funnel Chart

Sales pipeline stages.

---

10. Decomposition Tree Chart

Used for root cause analysis.

It automatically breaks down a measure into dimensions.

Example

Total Sales = 10M

Breakdown:

- Region
- Product
- Customer Segment

Suppose:
South Region has lowest sales.

Further breakdown:

- Product Category
- Salesperson

This helps identify the exact reason.

Advantages

- AI-based analysis
- Interactive drill-down
- Best for management dashboards

---

11. Types of DAX Functions

Main categories:

1. Aggregation
- SUM
- AVERAGE
- COUNT
2. Logical
- IF
- SWITCH
3. Filter
- FILTER
- ALL
- CALCULATE
4. Text
- LEFT
- RIGHT
- MID
5. Date & Time
- DATE
- YEAR
- MONTH
6. Time Intelligence
- TOTALYTD
- SAMEPERIODLASTYEAR
- DATESMTD

Time Intelligence Functions

Used for:

- YTD
- MTD
- YoY
- Previous month/year comparison

Example:

Sales LY =
CALCULATE(
    [Total Sales],

```
    SAMEPERIODLASTYEAR(DateTable[Date])
)
```

---

12. Month Wise Sales Calculation

Approach:

1. Create Date Table
2. Create relationship
3. Use Month column
4. Create measure

Total Sales = SUM(Sales[Amount])

Visual:

- Axis = Month
- Values = Total Sales

Best practice:
Use proper calendar table.

---

13. Handling Many-to-Many Relationship

Methods:

1. Bridge Table (Recommended)

Example:
Customer ↔ Bridge ↔ Product

---

2. CROSSFILTER

---

3. TREATAS

Best Practice

Use Star Schema and bridge table.

---

14. Joins in SQL

1. INNER JOIN

Returns matching records.

---

2. LEFT JOIN

All records from left table + matching right.

---

3. RIGHT JOIN

All records from right table + matching left.

---

4. FULL OUTER JOIN

All records from both tables.

---

5. CROSS JOIN

Cartesian product.

---

6. SELF JOIN

Table joined with itself.

---

15. DDL and DML Commands

DDL (Data Definition Language)

Used for structure.

Commands:

- CREATE
- ALTER
- DROP
- TRUNCATE

DML (Data Manipulation Language)

Used for data operations.

Commands:

- INSERT
- UPDATE
- DELETE
- MERGE

Round 2 Answers

1. Convert 20240629 into Date

SQL:

```
SELECT TO_DATE('20240629','YYYYMMDD');
```

Power BI DAX:

```
Date =
DATE(
LEFT([Column],4),
MID([Column],5,2),
RIGHT([Column],2)
)
```

---

2. Extract Last Name (Benjamin-Smith)

```
Last Name =
RIGHT(
    [Name],
    LEN([Name]) - FIND("-", [Name])
)
```

---

3. Ways of Division in DAX

1. Divide Operator

```
Profit % = [Profit] / [Sales]
```

---

2. DIVIDE Function (Recommended)

Handles divide-by-zero.

```
Profit % =
DIVIDE([Profit], [Sales], 0)
```

---

4. Extract Numbers from Column

Example:
AB123CD45

Power Query recommended.

DAX example:
Using SUBSTITUTE and PATH techniques.

Usually handled in Power Query with:

- Extract → Digits

---

5. Display Color Only for Highest Sales

```
Color Measure =
IF(
   [Total Sales] =
   MAXX(ALL(Product), [Total Sales]),
   "Green",
   "Gray"
)
```

Use conditional formatting.

---

6. How to Optimize Huge Reports

Methods:

1. Use Star Schema
2. Remove unnecessary columns
3. Use measures instead of calculated columns
4. Use Import mode where possible
5. Disable Auto Date/Time
6. Use Aggregations
7. Optimize DAX

8. Incremental Refresh
9. Reduce visual count
10. Use Performance Analyzer

---

7. How to Know Client Viewed Report

Methods:

1. Usage Metrics in Power BI Service
2. Audit Logs
3. Workspace monitoring
4. Activity Log API

Shows:

- Who viewed
- When viewed
- Frequency

---

8. DISTINCTCOUNT from Two Fact Tables

Use UNION.

```
Distinct Customers =
DISTINCTCOUNT(
    UNION(
        VALUES(Fact1[CustomerID]),
        VALUES(Fact2[CustomerID])
    )
)
```

Better:

```
Distinct Customers =
COUNTROWS(
    DISTINCT(
        UNION(
```

```
            VALUES(Fact1[CustomerID]),
            VALUES(Fact2[CustomerID])
        )
    )
)
```

---

9. How Did You Validate Report?

Validation steps:

1. Compare with source system
2. Validate totals with SQL
3. Cross-check KPIs
4. Validate filters/slicers
5. Test edge cases
6. User Acceptance Testing (UAT)
7. Performance testing

---

10. SUMMARIZE Function

Creates grouped summary table.

Example:

```
SUMMARIZE(
    Sales,
    Sales[Region],
    "Total Sales", SUM(Sales[Amount])
)
```

Equivalent to SQL GROUP BY.

---

11. 10 Developers and 500 Users

Use:

- Power BI Service
- Workspace Apps
- Security Groups
- RLS

Publish once and share through App.

Avoid individual sharing.

---

12. Inactive Relationship Handling

Use:

USERELATIONSHIP()

Example:

```
Sales by Ship Date =
CALCULATE(
    [Total Sales],
    USERELATIONSHIP(
        Sales[ShipDate],
        Date[Date]
    )
)
```

---

13. Refreshing 500 Million Rows in Import Mode

Use:

1. Incremental Refresh
2. Partitioning
3. Dataflows

4. Aggregation tables
5. Hybrid tables
6. Premium Capacity

Only latest data refreshes instead of full refresh.

---

14. Is This Measure Correct?

CALCULATE([Total Sales],[Total Sales] > 4000)

❌ Incorrect.

Reason:
Boolean expression cannot directly use measure like that.

Correct:

```
CALCULATE(
   [Total Sales],
   FILTER(
      ALL(Table),
      [Total Sales] > 4000
   )
)
```

---

15. Limitations of Direct Query Mode

1. Slower performance
2. Limited DAX functions
3. Dependency on source DB performance
4. Query timeout issues
5. Limited transformations
6. No full caching
7. Limited modeling features
8. High load on source system
9. Visual interaction slower
10. Maximum 1 million row return per query in some scenarios